

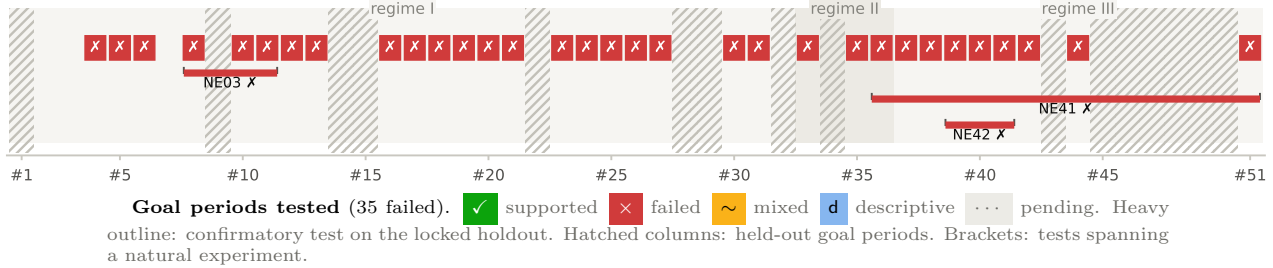
H45: Context homeostasis: each agent keeps its context near a set point

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 0.5/5 Usefulness 2/5

Bottom line. The room share of an agent's context is not regulated; it follows room traffic as passive accumulation predicts, and the scaffold's resets and windows, not the agents, bound it.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH166 claimed each agent holds the room-message share of its context near a set point: reading less per message when the room floods ($k^{-0.6}$ dilution as the homeostat), and overshooting, then relaxing, after an erasure. With DQ1's context ledger and measured prompt tokens we computed the share per model call on 32 non-holdout goal periods (#4–#51), plus four native tests: NE41's 21k forced erasures, the NE42 merge, #51's roster growth and NE03's chat fetch limit. Synthetic controller, passive and scaffold-capped agents on real call schedules came first. **Refuted.** The share follows inflow as passive accumulation predicts (regulation index ≤ 0.22 in 32/32 periods, pooled 0.02); floods move set points at 1.06 (NE42) and 0.90 (#51) times the passive slope. After an erasure agents talk *less* (0.74) and reply more per talk, with replies per call unchanged. Holdout script written, not run.

PHYSICAL MODEL

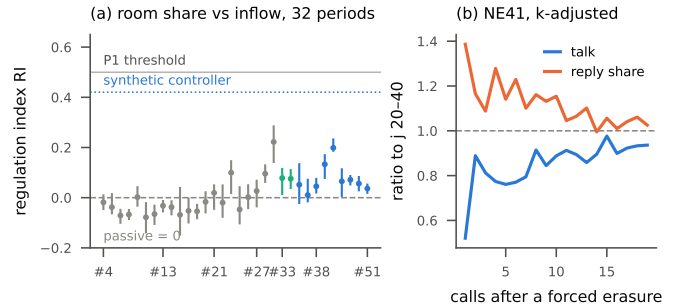
A set-point controller against passive accumulation (models 04, 05). In one context segment between resets, room content R_j (calibrated tokens of the new room items) and own content W_j (the rest of the prompt P_j) accumulate; the share is $s_j = R_j/P_j$. A controller would raise W or reset sooner when the share is high, and engage more when it is low. λ : room tokens per call; ε : elasticity of the segment share to λ ; η_W : own content's.

$$s_j = \frac{R_j}{R_j + W_j}, \quad R_j = \sum_{i \leq j} (a k_i + b c_i + e o_i)$$

$$\text{passive: } s_j = \frac{\lambda_j}{W_j + \lambda_j}, \quad W_j \text{ independent of } \lambda$$

$$\varepsilon = \left. \frac{\partial \ln \bar{s}}{\partial \ln \lambda} \right|_j = (1 - \bar{s})(1 - \eta_W), \quad \text{RI} = 1 - \frac{\varepsilon}{1 - \bar{s}} = \eta_W$$

$$P(\text{reply}) = 1 - \exp(-\theta_{\text{agent} \cdot \text{day}} k^{1-\beta})$$



Observables. (a) Regulation index per goal period (day-bootstrap CI; regime I gray, II aqua, III blue). Passive accumulation gives 0; the synthetic controller on the same schedules gives 0.42; the prediction needed ≥ 0.5 . Every period sits near passive. (b) NE41: after a forced erasure (41-call cap), k-adjusted talk propensity dips about 26% and recovers over about ten calls, while the share of talks that reply to pending messages rises. Replies per call stay flat (0.92).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Mostly a useful negative. Agents do not protect their context from a flooding room and do not “refill” it after an erasure. The share of a context window spent on chat is set by room traffic per call, the agent's own tool output and the scaffold's resets, so an operator who wants less chat in context must change the room (size, merges) or the scaffold (cadence, caps), not hope agents ration it. Merging rooms doubled the share. The 200-event cap almost never binds (0.05% of calls). Erasures cost about ten calls of reduced talk, not a catch-up burst. The engagement side is closer to context competition: replies fall as the agent's own work fills the context. Nothing here is a lever on the agents themselves; it is a design rule about where context composition is decided.

RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
P1 $RI \geq 0.5$ in $\geq 2/3$	0/32; CI upper < 0.5 in 32/32; pooled 0.02	failed
P3 share band (CV ratio ≤ 0.5)	0.84–1.32, median 0.99	failed
P2 reset sooner when flooded	3/9 regime III; #51 -0.04 (controller -0.10)	failed (weak)
P4 dilution $\beta \in [0.4, 0.9]$	21/32; pooled 0.82 (III 0.98); H18 ρ 0.71	narrowly failed
P5 $\gamma_R < 0$, $\gamma_W > 0$ (pooled III)	$\gamma_R +0.08$, $\gamma_W -0.10$ (42k talks)	failed
P6 post-erasure over-shoot	0/9; pooled talk 0.74 [0.69, 0.80]	reversed
P7 lab sets the set point	lab η^2 0.14, p 0.41	failed
P8/P9 no truncation; cap rare	P doubles in-segment; cap 0.05%	passed
NE42 / #51 leak ≤ 0.3	1.06 [0.95, 1.11] / 0.90 [0.77, 0.99]	failed
NE03 chat ceiling ap-pears	flat before and after (8–10k to-kens)	failed

GOAL PERIODS IN DETAIL

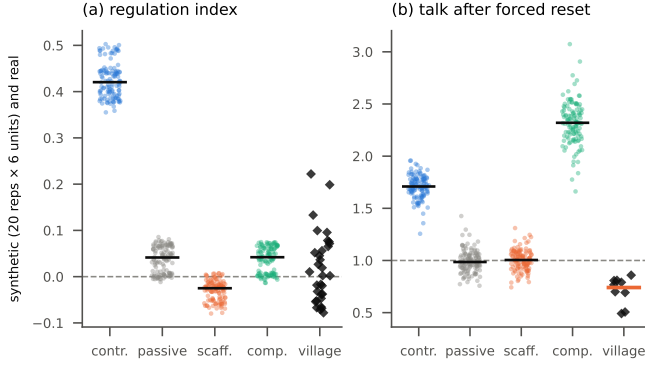
G04 $\times$ failed (RI -0.02 [-0.05, 0.01])
G05 $\times$ failed (RI -0.04 [-0.06, 0.01])
G06 $\times$ failed (RI -0.07 [-0.10, -0.05])
G08 $\times$ failed (RI -0.07 [-0.08, -0.05])
G10 $\times$ failed (RI 0.00 [-0.03, 0.04])
G11 $\times$ failed (RI -0.08 [-0.14, -0.02])
G12 $\times$ failed (RI -0.07 [-0.10, -0.03])
G13 $\times$ failed (RI -0.03 [-0.05, -0.01])
G16 $\times$ failed (RI -0.04 [-0.07, -0.01])
G17 $\times$ failed (RI -0.07 [-0.15, 0.04])

+25 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).



Second figure. Synthetic validation on real schedules (6 units, 20 replicates; dots) with the village (diamonds: per-period estimates; orange bar: pooled NE41). The regulation index separates a controller from passive, scaffold and competition agents at village counts. The village falls in the passive column. After erasures it talks less than any synthetic class.

CAVEATS

Room content R is a lower bound (`search_history` results uncounted; reset snapshot ≈ 0). Shared error in R pulls ε toward passive, so a weak controller (RI < 0.15) would be hard to see. Dense prompt tokens exist only for Anthropic and Google. One DQ2 reply parent per message steepens β . The engagement-on-context test is not identified per period and is confounded with task phase. NE42 is goal-confounded. Holdout not used.

ROUND 2

- **Essence.** whether agents manage their information intake at all, or the scaffold does it for them.
- **H45-R1** Context competition (HH158): does reply engagement fall with own-content fill once DQ3 behavior states control task phase?
- **H45-R2** Cadence as the agents' only intake lever: do call intervals shorten when the room's per-minute rate rises?
- **H45-R3** What fills the post-erasure talk dip?
- **H45-R4** Run `confirm.py` on the holdout (#45–#50, 51m, NE22) after the reuse disclosures.

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